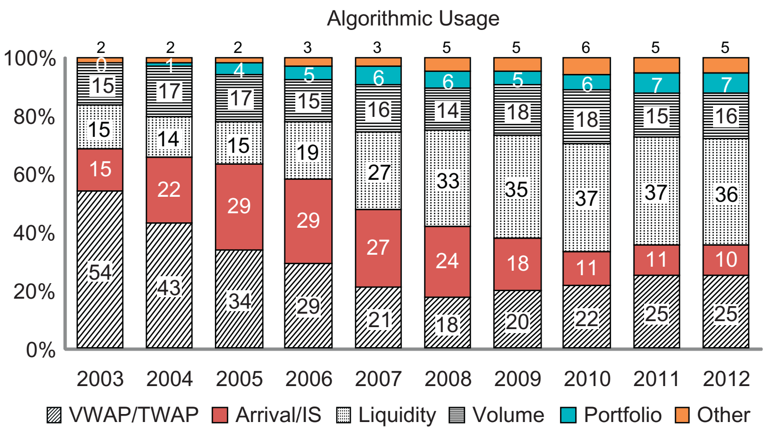
arbitrage. Black box trading strategies and the corresponding mathematics are discussed in detail in Chapter 13.

ALGORITHMIC TRADING TRENDS

Algorithmic usage patterns have also changed with the evolution of trading algorithms. In the beginning, algorithmic trading was mostly dominated by VWAP/TWAP trading that utilized a schedule to execute orders. The advantage: investors acquired a sound performance benchmark, VWAP, to use for comparison purposes. The improvement in algorithms and their ability to source liquidity and manage micro order placement strategies more efficiently led the way for price-based algorithms such as arrival price, implementation shortfall, and the Aggressive in the Money (AIM) and Passive in the Money (PIM) tactics. During the financial crisis, investors were more concerned with urgent trading and sourcing liquidity and many turned to "Liquidity Seeking" algorithms to avoid the high market exposure present during these times. The financial crisis resulted in higher market fragmentation owing to numerous venues and pricing strategies and a proliferation of dark pools. However, the industry is highly resourceful and quick to adapt. Firms developed internal crossing networks to match orders before being exposed to markets, providing cost benefits to investors, and incorporating much of the pricing logic and smart order capabilities into the liquidity seeking algorithms. Thus usage in these algos has remained relatively constant. Currently, liquidity seeking algos account for 36% of volumes, VWAP/TWAP 25%, volume 16%, arrival price/IS 10%, and portfolio algos 7%. This is shown in Figure 1.12.



■ **Figure 1.12** Algorithmic Usage (values in the graph may not add to 100 due to rounding).